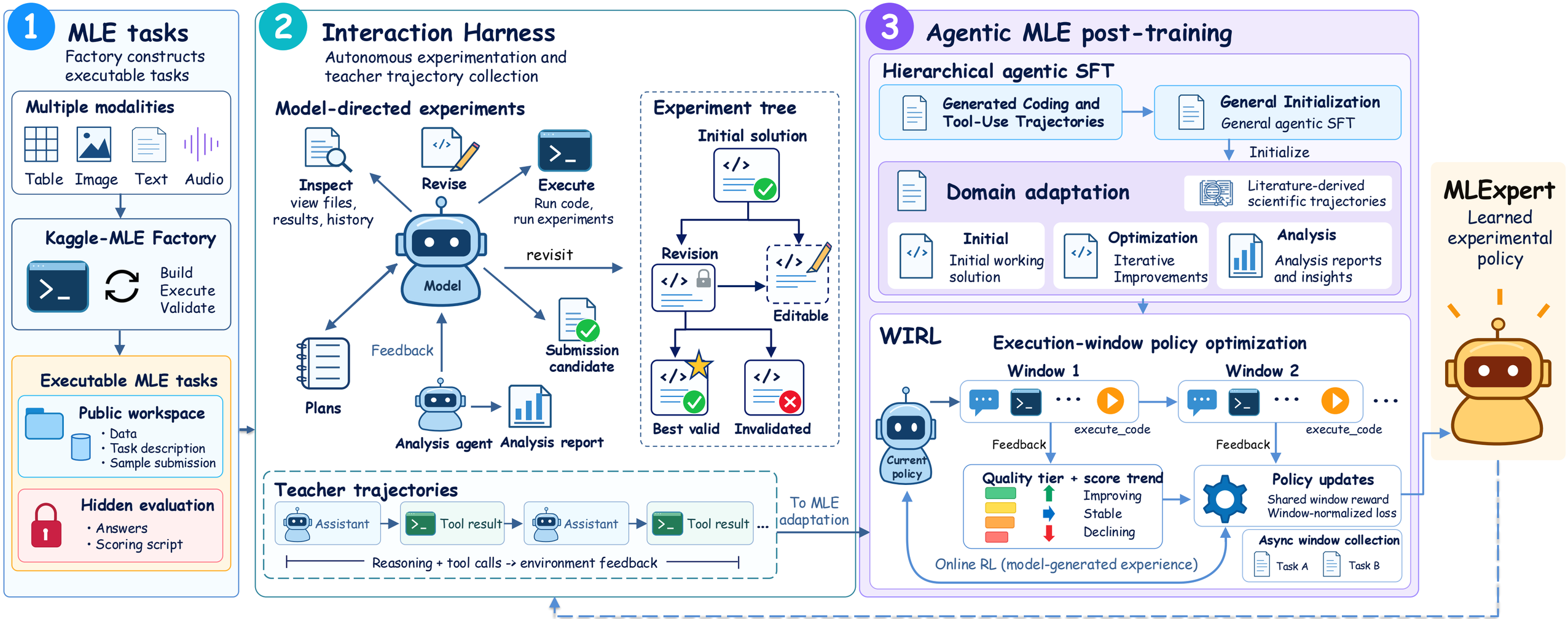

1
MLE tasks
Factory constructs executable tasks
Multiple modalities
Table
Image
Text
Audio
Kaggle-MLE Factory
Build
Execute
Validate
Executable MLE tasks
Public workspace
• Data
• Task description
• Sample submission
Hidden evaluation
• Answers
• Scoring script
2
Interaction Harness
Autonomous experimentation and teacher trajectory collection
Model-directed experiments
Inspect
view files, results, history
Revise
Execute
Run code, run experiments
Model
revisit
Submission candidate
Feedback
Plans
Analysis agent
Analysis report
Experiment tree
Initial solution
Revision
Editable
Best valid
Invalidated
Teacher trajectories
Assistant
Tool result
Assistant
Tool result
...
Reasoning + tool calls -> environment feedback
To MLE adaptation
3
Agentic MLE post-training
Hierarchical agentic SFT
Generated Coding and Tool-Use Trajectories
General Initialization
General agentic SFT
Initialize
Domain adaptation
Literature-derived scientific trajectories
Initial
Initial working solution
Optimization
Iterative Improvements
Analysis
Analysis reports and insights
WIRL
Execution-window policy optimization
Window 1
Window 2
execute_code
execute_code
Feedback
Feedback
Current policy
Quality tier + score trend
Improving
Stable
Declining
Policy updates
Shared window reward
Window-normalized loss
Async window collection
Task A
Task B
Online RL (model-generated experience)
MLExpert
Learned experimental policy